BIG DATA ANALYTICS(BCS714D) Question Bank

1. Compare Traditional Business Intelligence (BI) with Big Data Analytics. Highlight key differences with examples.
2. Define Big Data. Explain its characteristics (Volume, Velocity, Variety, Veracity, etc.) with suitable examples.
1st 3. Explain the importance of Big Data Analytics and list some top analytical tools used. 31 &38
1st 4. Discuss the classification of analytics (Descriptive, Predictive, Prescriptive) with use cases. 28
1 5. Compare a typical Data Warehouse environment with a Hadoop environment.
6. Describe the core components of Hadoop: HDFS and MapReduce.
7. Write a note on the Hadoop ecosystem. Explain the role of Hive, Pig, Sqoop, and HBase.
8. Explain any two use cases where Hadoop is effectively applied. Explain Hadoop Distributed File System (HDFS) architecture and its working.
9. Explain the data processing flow in Hadoop from data ingestion to final output using HDFS, YARN, and MapReduce.
10. Describe the phases of a MapReduce program: Mapper, Reducer, Combiner, and Partitioner.
11. Why is MongoDB preferred over traditional RDBMS in Big Data applications?
12. Differentiate terms used in RDBMS and MongoDB with examples.
13. Explain the different data types supported in MongoDB.
14. Write short notes on the MongoDB Query Language (MQL) with examples of basic queries.